# George Gu

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## Education

**University of Michigan** May 2028
***M.S. in Computer Science, Machine Learning (Expected)*** *Ann Arbor, MI*

**University of Michigan** May 2027
***Bachelor of Science in Engineering, Computer Science*** *Ann Arbor, MI*

- **GPA**: 3.81/4.00
- **Coursework**: Machine Learning, Artificial Intelligence, Reinforcement Learning, Operating Systems, Theory of Computation, Computer Organization, Web Systems, Computer Security

## Experience

**Capital One** June – August 2026
*Technology Intern* *Richmond, VA*

- Architected data pipelines via scheduled **AWS Lambda** workers to ingest daily snapshots from internal asset-management systems, yielding clean tabular datasets to train predictive models on application lifecycle and business criticality.
- Engineered Python **REST APIs** orchestrating data-attestation and compliance workflows, securing them with parameterized PostgreSQL queries and role-based access control across corporate domains and org hierarchies.
- Authored **20+** automated integration tests covering mock database transactions, cross-region failures, and network-timeout edge cases, lifting repository-wide coverage to **91%** for the compliance service.

**BoilerVault** January 2026 – Present
*Founding Engineer (Contract)* *West Lafayette, IN*

- Built a multi-tenant operations platform with FastAPI, PostgreSQL, and Next.js/TypeScript for a **3-campus** student storage business, implementing JWT auth and 20+ REST endpoints validated by **140+ tests.**
- Reconciled out-of-order payment events across **4 Stripe accounts** and **3 WordPress booking sites** via idempotent webhook pipelines, automating customer, order, and billing creation with 0 manual entry across **$130K+** in gross revenue.
- Migrated **2,000+** legacy records via three-tier matching (exact email, rapidfuzz fuzzy name, Stripe charge fallback), replacing a failing Zapier workflow and deploying on Railway/Vercel as the production system of record.

**Nexteer Automotive** May – August 2025
*Software Engineering Intern* *Saginaw, MI*

- Developed an **LLM-powered** IDE extension in Python and TypeScript that parsed **300+** internal engineering guidelines and automated compliance checks across C and H files, integrating Azure AI and Copilot APIs.
- Designed prompt pipelines and few-shot strategies to extract rules from unstructured documentation, tuning the system to **95% violation-detection accuracy** and saving engineers an estimated 8 hours of review per week.
- Deployed the tool across **26 sites** and integrated it with internal dev pipelines, cutting manual code-review time by **87.5%** for safety-critical embedded steering systems.

## Projects

**Image Classification & Transfer Learning** January – April 2026
*Machine Learning Developer* *Ann Arbor, MI*

- Implemented a **Vision Transformer** from scratch in **PyTorch**, building patch-to-token projection, a learnable classification token, sinusoidal positional embeddings, and multi-head self-attention encoder blocks.
- Engineered a 2-stage pretrain/fine-tune pipeline, sweeping augmentation intensity across 4 levels and tuning early-stopping patience on validation AUROC to correct overfitting, reaching **0.93 accuracy** and **0.97 AUROC** on validation.

**Scalable Search Engine** January – April 2026
*Systems Developer* *Ann Arbor, MI*

- Built a multi-stage MapReduce pipeline in Python to construct an inverted index over **3,000+ documents**, computing TF-IDF scores and document-normalization factors across a parallel, multi-job architecture.
- Implemented a REST API index server in Flask serving **3 partitioned inverted index segments**, integrating PageRank-weighted cosine similarity scoring to balance TF-IDF with link authority when ranking results.

**Resume Screener** January – May 2025
*Machine Learning Developer* *Ann Arbor, MI*

- Built an **NLP** pipeline over a labeled dataset of **1,000+ resumes and job postings**, applying tokenization, TF-IDF vectorization, and feature engineering for resume-to-posting matching.
- Designed **scikit-learn** and **TensorFlow** models to rank resumes against job postings, using sentence-transformer embeddings and **RAG** retrieval over job descriptions, deployed via Streamlit to **100+ users.**

## Skills/Additional

**Languages**: Python, SQL, C/C++, Java, JavaScript, TypeScript, HTML, CSS
**Frameworks/Developer Tools**: PyTorch, TensorFlow, Git, Docker, Kubernetes, Unix, Node.js, React.js, Flask, PostgreSQL
**Interests**: Philosophy and Metaphysics, Game Design, Bouldering, Skateboarding, Golf, Guitar, Calisthenics

## Publication

**Gu G**, Pei H, Zhou A, Fan B, Zhou H, Choi A, Huang Z, "A Comprehensive Study of Historical Detection Data for Pathogen Isolates from U. S. Cattle", 2023, *Antibiotics*, 12, 1509.
doi: https://doi.org/10.3390/antibiotics12101509